Java Enterprise, Spring Ecosystem & Web Technologies

Comprehensive 100-Question Assessment Bank

Topics: Core Java, JDBC, Servlets, JSP, ORM, Hibernate, Spring IoC, Spring AOP, Spring Boot, Spring Data JPA, REST APIs, Maven, Postman, STS & Servers

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Core Java Fundamentals

Q1. Which of the following features was introduced in Java 8 to support functional programming?

(A) Generics
(B) Lambda Expressions
(C) Annotations
(D) Reflection API

Q2. What is the contract between `equals()` and `hashCode()` methods in Java?

(A) If two objects are equal, they must have different hash codes
(B) If two objects have the same hash code, they must be equal
(C) If two objects are equal, they must have the same hash code
(D) No relation exists between them

Q3. Which memory area in JVM stores class structures such as runtime constant pool, field, and method data?

(A) Java Heap
(B) Metaspace
(C) Stack Memory
(D) Program Counter Register

Q4. What happens when a thread encounters an unhandled exception in Java?

(A) The JVM crashes immediately
(B) The thread terminates, and its stack trace is printed
(C) The exception is automatically passed to garbage collection
(D) All running threads are terminated

Q5. Which interface must be implemented by a class whose instances are intended to be executed by a thread?

(A) `java.lang.Callable`
(B) `java.lang.Runnable`
(C) `java.lang.Executor`
(D) `java.lang.Thread`

Q6. Which of the following collections guarantees insertion order and allows duplicate elements?

(A) `HashSet`
(B) `TreeSet`
(C) `ArrayList`
(D) `HashMap`

Q7. What is the main purpose of the `volatile` keyword in Java?

(A) Make a variable immutable
(B) Prevent thread synchronization
(C) Ensure visibility of changes across threads directly to main memory
(D) Serialization of objects

Q8. Which method in `Optional` class returns the value if present, otherwise returns a default value?

(A) `get()`
(B) `orElse()`
(C) `isPresent()`
(D) `ifPresent()`

JDBC (Java Database Connectivity)

Q9. Which JDBC interface is used to execute parameterized SQL queries efficiently multiple times?

(A) `Statement`
(B) `PreparedStatement`
(C) `CallableStatement`
(D) `Driver`

Q10. Which method of `DriverManager` is called to register a database driver class explicitly in older JDBC versions?

(A) `Class.forName()`

(B) `Driver.register()`

(C) `Connection.getDriver()`

(D) `DriverManager.load()`

Q11. What is the default auto-commit mode for a newly created JDBC `Connection`?

(A) `false`

(B) `true`

(C) Undefined

(D) Read-only

Q12. Which interface is preferred for database connection pooling in enterprise production applications?

(A) `java.sql.DriverManager`

(B) `javax.sql.DataSource`

(C) `java.sql.Connection`

(D) `java.sql.Driver`

Q13. How do you process a stored procedure returning multiple result sets using JDBC?

(A) `Statement`

(B) `PreparedStatement`

(C) `CallableStatement`

(D) `ResultSetMetaData`

Q14. Which `ResultSet` concurrency type allows modifying database data directly through the result set object?

(A) `CONCUR_READ_ONLY`

(B) `CONCUR_UPDATABLE`

(C) `TYPE_SCROLL_INSENSITIVE`

(D) `TYPE_FORWARD_ONLY`

Q15. Which exception is thrown when a database access error or other JDBC errors occur?

(A) `IOException`

(B) `DatabaseException`

(C) `SQLException`

(D) `DataAccessException`

Servlets Technology

Q16. What is the lifecycle order of a Java Servlet?

(A) `init()` → `service()` → `destroy()`

(B) `service()` → `init()` → `destroy()`

(C) `init()` → `destroy()` → `service()`

(D) `create()` → `init()` → `service()`

Q17. Which method in `HttpServlet` handles HTTP POST requests?

(A) `doPost(ServletRequest, ServletResponse)`

(B) `doPost(HttpServletRequest, HttpServletRespon`

(C) `handlePost(HttpServletRequest, HttpServletResponse)`

(D) `servicePost(HttpServletRequest, HttpServletResponse)`

Q18. Which object is shared across the entire web application and accessible to all servlets?

(A) `ServletConfig`

(B) `ServletContext`

(C) `HttpSession`

(D) `HttpServletRequest`

Q19. How can parameters defined in `web.xml` under a specific `<servlet>` tag be retrieved?

(A) `getServletContext().getInitParameter()`

(B) `getServletConfig().getInitParameter()`

(C) `request.getParameter()`

(D) `session.getAttribute()`

Q20. Which method is used to redirect a client request to another URL on a different server?

(A) `RequestDispatcher.forward()`

(B) `HttpServletResponse.sendRedirect()`

(C) `RequestDispatcher.include()`

(D) `ServletContext.dispatch()`

Q21. Annotation introduced in Servlet 3.0 to declare servlets without using `web.xml`:

(A) `@Servlet`

(B) `@WebServlet`

(C) `@WebComponent`

(D) `@Controller`

Q22. What is the scope of an attribute stored in `HttpSession`?

(A) Accessible across all users
(B) Accessible only during a single HTTP request
(C) Accessible across multiple requests from the same client session
(D) Accessible until JVM restarts

JSP (JavaServer Pages)

Q23. What is the correct syntax for a JSP Scriptlet tag?

(A) `<%! ... %>`
(B) `<%= ... %>`
(C) `<% ... %>`
(D) `<%-- ... --%>`

Q24. Which JSP implicit object corresponds to the `ServletConfig` instance?

(A) `application`
(B) `config`
(C) `pageContext`
(D) `session`

Q25. What is the purpose of the JSP declaration tag `<%! ... %>`?

(A) To print output directly to the HTTP response stream
(B) To declare class-level variables and methods
(C) To import external Java packages
(D) To execute Java statements within `_jspService()`

Q26. Which JSTL library prefix is commonly used for core actions like iteration and conditional branching?

(A) `fmt`
(B) `fn`
(C) `c`
(D) `sql`

Q27. Which Directive is used to import packages into a JSP page?

(A) `<%@ include %>`
(B) `<%@ page import="..." %>`
(C) `<%@ taglib %>`
(D) `<%@ useBean %>`

Q28. What is the default scope of a `<jsp:useBean>` tag if not specified?

(A) `request`
(B) `session`
(C) `page`
(D) `application`

Q29. Expression Language (EL) syntax in JSP begins with which character?

(A) `#{...}`
(B) `${...}`
(C) `@{...}`
(D) `*{...}`

ORM & Hibernate Concepts

Q30. What does ORM stand for in Java enterprise development?

(A) Object Relational Mapping
(B) Object Resource Management
(C) Operational Relational Model
(D) Optimized Repository Mapping

Q31. Which Hibernate interface is a factory for `Session` objects and thread-safe?

(A) `Session`
(B) `SessionFactory`
(C) `Transaction`
(D) `Query`

Q32. What state is an entity in when it is instantiated with `new` but not yet associated with a Hibernate Session?

(A) Persistent
(B) Detached
(C) Transient
(D) Removed

Q33. Which annotation marks a class as a persistent entity in Hibernate/JPA?

(A) `@Table`
(B) `@Entity`
(C) `@Persistent`
(D) `@Model`

Q34. Which annotation specifies the primary key of an entity?

(A) `@PrimaryKey`

(B) `@Id`

(C) `@Key`

(D) `@Column(primary=true)`

Q35. What is the purpose of Hibernate's First-Level Cache?

(A) Caches query results across applications

(B) Mandatory cache bound to the current Session

(C) Optional cache configured via EHCache

(D) Database server cache

Q36. Which method transitions an entity from transient to persistent state in Hibernate?

(A) `session.save()` / `session.persist()`

(B) `session.merge()`

(C) `session.update()`

(D) `session.detach()`

Q37. What exception occurs when lazy loading uninitialized collections outside an active session?

(A) `NullPointerException`

(B) `LazyInitializationException`

(C) `EntityNotFoundException`

(D) `SessionClosedException`

Q38. Which inheritance strategy produces a single table containing columns for all subclass properties?

(A) `SINGLE_TABLE`

(B) `JOINED`

(C) `TABLE_PER_CLASS`

(D) `MAPPED_SUPERCLASS`

Q39. How do you represent a One-to-Many entity association with owner side configuration?

(A) `@OneToMany(mappedBy="...")`

(B) `@ManyToOne` on inverse side

(C) Both A and B

(D) `@ManyToMany`

Spring Framework Core & IoC

Q40. What fundamental software engineering principle underlies Spring IoC Container?

(A) Factory Pattern

(B) Inversion of Control / Dependency Injection

(C) Observer Pattern

(D) Singleton Pattern

Q41. Which Spring annotation is a stereotype used to mark a class as a business service component?

(A) `@Repository`

(B) `@Component`

(C) `@Service`

(D) `@Bean`

Q42. What is the default bean scope in the Spring IoC container?

(A) `prototype`

(B) `request`

(C) `singleton`

(D) `session`

Q43. Which annotation autowires dependencies by type in Spring?

(A) `@Inject`

(B) `@Autowired`

(C) `@Resource`

(D) `@Value`

Q44. When multiple candidate beans exist for injection, which annotation eliminates ambiguity by name?

(A) `@Primary`

(B) `@Qualifier`

(C) `@Name`

(D) `@Scope`

Q45. Which interface represents the core Spring IoC container interface with advanced enterprise features?

(A) `BeanFactory`

(B) `ApplicationContext`

(C) `BeanDefinition`

(D) `ObjectFactory`

Q46. What method is called immediately after bean property population if using lifecycle annotations?

(A) `@PreDestroy`
(B) `@PostConstruct`
(C) `init()`
(D) `setup()`

Q47. Which bean scope creates a new bean instance for every request made to the Spring container?

(A) `singleton`
(B) `prototype`
(C) `application`
(D) `websocket`

Spring AOP (Aspect-Oriented Programming)

Q48. What is an "Aspect" in Spring AOP?

(A) A method that executes database transactions
(B) A module encapsulating cross-cutting concerns
(C) An event listener
(D) A REST controller

Q49. Which AOP advice type executes both before and after a target method execution?

(A) `@Before`
(B) `@After`
(C) `@Around`
(D) `@AfterReturning`

Q50. In Spring AOP, what term defines a point during application execution where an aspect can be plugged in?

(A) Pointcut
(B) Join Point
(C) Target Object
(D) Advice

Q51. What is a "Pointcut"?

(A) The code executed by an Aspect
(B) A predicate that matches Join Points
(C) The proxy class generated by Spring
(D) A transaction boundary

Q52. Which advice executes only if a method completes without throwing any exceptions?

(A) `@AfterThrowing`
(B) `@AfterReturning`
(C) `@After`
(D) `@Around`

Q53. Spring AOP uses which dynamic proxy mechanisms by default when interfaces are present?

(A) CGLIB proxies
(B) JDK Dynamic Proxies
(C) AspectJ compiler
(D) Javassist

Spring Boot

Q54. What is the main purpose of Spring Boot Auto-Configuration?

(A) To eliminate XML configurations entirely
(B) Automatically configure Spring beans based on classpath dependencies
(C) Generate database tables automatically
(D) Provide security tokens

Q55. Which combination of annotations is included inside `@SpringBootApplication`?

(A) `@Configuration`, `@EnableAutoConfiguration`, `@ComponentScan`
(B) `@Controller`, `@Service`, `@Repository`
(C) `@EnableJpaRepositories`, `@Entity`, `@Bean`
(D) `@SpringBootConfiguration`, `@WebMvcTest`, `@Autowired`

Q56. What embedded server is default in a standard `spring-boot-starter-web` application?

(A) Jetty
(B) Undertow
(C) Apache Tomcat
(D) GlassFish

Q57. Which Spring Boot file format is widely used for hierarchical configuration properties?

(A) `application.xml`

(B) `application.yml` or `application.properties`

(C) `spring.config`

(D) `bootstrap.json`

Q58. Which dependency starter is required for building Web applications including RESTful APIs in Spring Boot?

(A) `spring-boot-starter-data-jpa`

(B) `spring-boot-starter-web`

(C) `spring-boot-starter-test`

(D) `spring-boot-starter-actuator`

Q59. What module does Spring Boot provide to monitor and manage metrics, health, and environment details?

(A) Spring Security

(B) Spring Boot Actuator

(C) Spring DevTools

(D) Spring Cloud Config

Q60. How do you override default properties (e.g., server port) in Spring Boot?

(A) Modify `web.xml`

(B) Set `server.port=8081` in `application.properties`

(C) Write explicit Java code in main class

(D) Pass properties via database table

Q61. Which interface can be implemented to execute code blocks right after Spring Boot startup completes?

(A) `CommandLineRunner` / `ApplicationRunner`

(B) `ServletInitializer`

(C) `BeanFactoryPostProcessor`

(D) `SpringApplicationBuilder`

Spring Data JPA

Q62. Which repository interface in Spring Data JPA provides pagination and sorting capabilities out of the box?

(A) `CrudRepository`

(B) `PagingAndSortingRepository`

(C) `JpaRepository`

(D) Both B and C

Q63. How does Spring Data JPA auto-generate queries without writing explicit SQL/JPQL?

(A) Through XML mapping

(B) By parsing method names following naming conventions

(C) Via external stored procedures

(D) By reading database metadata

Q64. Which annotation allows writing explicit custom queries in Spring Data JPA interfaces?

(A) `@SQL`

(B) `@Query`

(C) `@NativeQuery`

(D) `@Finder`

Q65. What annotation is mandatory on custom queries performing UPDATE or DELETE operations in Spring Data JPA?

(A) `@Transactional`

(B) `@Modifying`

(C) `@Mutate`

(D) `@Update`

Q66. How do you enable automatic audit field population (e.g., `@CreatedDate`) in Spring Data JPA?

(A) `@EnableJpaAuditing`

(B) `@EnableTransactionManagement`

(C) `@EnableJpaRepositories`

(D) `@Auditable`

Q67. What return type is used in Spring Data JPA to wrap query results and avoid `NullPointerException`?

(A) `List`

(B) `Optional<T>`

(C) `CompletableFuture`

(D) `Set`

REST APIs & Spring MVC

Q68. Which annotation combines `@Controller` and `@ResponseBody` in Spring?

(A) `@WebController`
(B) `@RestController`
(C) `@APIController`
(D) `@ServiceController`

Q69. Which HTTP method is idempotent and intended to update existing resources completely?

(A) POST
(B) PUT
(C) PATCH
(D) DELETE

Q70. Which annotation extracts URI template variables from request endpoints (e.g., `/users/{id}`)?

(A) `@RequestParam`
(B) `@PathVariable`
(C) `@RequestBody`
(D) `@RequestHeader`

Q71. Which HTTP status code represents "201"?

(A) OK
(B) Accepted
(C) Created
(D) No Content

Q72. Which annotation binds HTTP request JSON body to a Java DTO parameter in controllers?

(A) `@ModelAttribute`
(B) `@RequestBody`
(C) `@ResponseBody`
(D) `@RequestPayload`

Q73. Which HTTP status code indicates a client requested a resource that does not exist?

(A) 400 Bad Request
(B) 401 Unauthorized
(C) 403 Forbidden
(D) 404 Not Found

Q74. How do you handle exceptions globally across controllers in Spring Boot?

(A) Using `@ControllerAdvice` / `@RestControllerAdvice`
(B) Custom `Filter`
(C) Standard Java `try-catch` blocks
(D) Overriding `HttpServlet`

Q75. What header field is used by clients to request specific payload format (e.g., `application/json`)?

(A) `Content-Type`
(B) `Accept`
(C) `Authorization`
(D) `User-Agent`

Maven Build Tool

Q76. What is the core configuration XML file used in Apache Maven?

(A) `build.xml`
(B) `settings.xml`
(C) `pom.xml`
(D) `maven.xml`

Q77. What does POM stand for in Maven?

(A) Project Object Model
(B) Program Optimization Management
(C) Packaging Object Module
(D) Process Operational Mapping

Q78. Which phase in Maven default build lifecycle compiles source code?

(A) `validate`
(B) `compile`
(C) `test`
(D) `package`

Q79. Where does Maven store downloaded dependencies locally by default?

(A) `/usr/local/maven`
(B) `~/.m2/repository`
(C) `C:\Program Files\Maven`
(D) `./target/dependency`

Q80. Which command cleans target directory and creates executable JAR artifact?

(A) `mvn clean install`
(B) `mvn compile package`
(C) `mvn build`

(D) `mvn deploy`

Q81. What dependency scope means a JAR is provided by the application server (e.g., Servlet API)?

(A) `compile`
(B) `provided`
(C) `runtime`
(D) `test`

Postman & API Testing

Q82. In Postman, which tab is used to pass key-value parameters appended to URL query string?

(A) Headers
(B) Params
(C) Body
(D) Pre-request Script

Q83. Which authentication type in Postman accepts token value directly in the request header?

(A) Basic Auth
(B) Bearer Token
(C) Digest Auth
(D) OAuth 1.0

Q84. What Postman feature enables sharing reusable base URLs across different environments (Dev, QA, Prod)?

(A) Postman Collections
(B) Environment Variables
(C) Monitors
(D) Mock Servers

Q85. Which tab in Postman allows writing JavaScript assertions to test response HTTP status code?

(A) Pre-request Script
(B) Tests
(C) Body
(D) Authorization

Q86. Which body format in Postman is used to submit standard HTML form submissions (`multipart/form-data`)?

(A) `raw (JSON)`
(B) `form-data`
(C) `x-www-form-urlencoded`
(D) `binary`

Spring Tool Suite (STS) & IDEs

Q87. What is Spring Tool Suite (STS)?

(A) An application server
(B) An Eclipse-based IDE customized for Spring Application development
(C) A testing library
(D) A database GUI tool

Q88. Which feature in STS/Eclipse allows rapidly generating getters, setters, and constructors?

(A) Source Code Generator Menu
(B) Spring Boot Initializr wizard
(C) Spring Live Beans
(D) Refactoring Tools

Q89. Which tool integrated inside STS allows bootstrap creating new Spring Boot projects easily?

(A) Maven CLI
(B) Spring Initializr Integration
(C) Spring Boot Dashboard
(D) Both B and C

Q90. How do you run a Spring Boot application inside STS IDE?

(A) Right-click Project $\rightarrow$ Run As $\rightarrow$ Spring Boot App
(B) Run `tomcat:run`
(C) Deploy WAR manually
(D) Execute `java -jar` command in external terminal

Web Servers & Application Servers

Q91. What is the main difference between Web Server and Application Server?

(A) Web servers process static HTTP; App servers support enterprise EJB/JMS/Full stack specs
(B) App servers cannot serve HTTP content
(C) Web servers can run EJB containers
(D) No difference

Q92. Apache Tomcat is primarily classified as what type of container?

(A) Full Application Server
(B) Servlet/JSP Web Container
(C) Database Engine
(D) Message Broker

Q93. What file in Apache Tomcat configures global server settings and ports?

(A) `web.xml`
(B) `server.xml`
(C) `context.xml`
(D) `tomcat-users.xml`

Q94. What is the default HTTP connector port configured in Apache Tomcat?

(A) 80
(B) 8080
(C) 443
(D) 9090

Q95. Which deployment archive format is used to deploy enterprise applications containing web and EJB modules?

(A) JAR
(B) WAR
(C) EAR
(D) ZIP

Advanced Mixed Concepts

Q96. In Spring Transaction Management, what does `@Transactional(readOnly = true)` optimize?

(A) Database locks and Hibernate flush modes for read operations
(B) Prevents writing to memory
(C) Disables network connections
(D) Automatically updates records

Q97. What HTTP response code is returned when payload validation (`@Valid`) fails in Spring Boot?

(A) 500 Internal Server Error
(B) 400 Bad Request
(C) 422 Unprocessable Entity
(D) 403 Forbidden

Q98. Which annotation is used in Spring Boot to map properties from `application.yml` to Java objects safely?

(A) `@Value`
(B) `@ConfigurationProperties`
(C) `@PropertySource`
(D) `@Bean`

Q99. Which annotation handles CORS (Cross-Origin Resource Sharing) requests in Spring REST Controllers?

(A) `@CrossOrigin`
(B) `@AllowOrigin`
(C) `@CORSEnable`
(D) `@SecurityHeader`

Q100. What is N+1 Select Problem in ORM/Hibernate?

(A) Fetching 1 parent record causes N separate SQL queries for child collections
(B) Inserting 1 record requires N table locks
(C) Connecting 1 user creates N threads
(D) None of the above

Q101. Which fetch type strategy avoids loading child collection until explicitly accessed?

(A) `FetchType.EAGER`
(B) `FetchType.LAZY`
(C) `FetchType.MERGE`
(D) `FetchType.BATCH`

Q102. How do you solve the N+1 Select Problem in JPA Query?

(A) Use `JOIN FETCH` in JPQL query
(B) Use Entity Graphs
(C) Set Batch Size

(D) All of the above

Q103. Which annotation is used to inject values from `application.properties` into single fields?

(A) `@Autowired`

(B) `@Value`

(C) `@Inject`

(D) `@Property`

Q104. What is the default transaction propagation setting in Spring `@Transactional`?

(A) `REQUIRED`

(B) `REQUIRES_NEW`

(C) `SUPPORTS`

(D) `NEVER`

Q105. Which design pattern is heavily used by Spring Framework for managing template operations like `JdbcTemplate`?

(A) Strategy Pattern

(B) Template Method Pattern

(C) Adapter Pattern

(D) Builder Pattern

Q106. How do you configure Spring Security to secure REST API endpoints in modern Spring Boot?

(A) Define a `SecurityFilterChain` bean

(B) Extend `WebSecurityConfigurerAdapter`

(C) Add `security.xml` file

(D) Configure `web.xml`

Q107. What does JSON stand for in web API communication?

(A) Java Standard Object Notation

(B) JavaScript Object Notation

(C) Java Structured Object Network

(D) JavaScript Oriented Network

Q108. Which HTTP header is commonly used to send JWT Bearer tokens for API authentication?

(A) `Content-Type`

(B) `Authorization`

(C) `X-API-Key`

(D) `Accept-Encoding`

Q109. Which Maven plugin compiles Java sources into class files?

(A) `maven-surefire-plugin`

(B) `maven-compiler-plugin`

(C) `maven-jar-plugin`

(D) `spring-boot-maven-plugin`

Q110. What is Jackson library used for in Spring Web applications?

(A) Database connectivity

(B) Object-to-JSON serialization and deserialization

(C) Security authentication

(D) Logging application events

Q111. Which annotation is used on JPA entities to implement optimistic locking?

(A) `@Lock`

(B) `@Version`

(C) `@Optimistic`

(D) `@Synchronized`

Q112. In Spring Boot, how can you package an application as a runnable WAR file instead of standard JAR?

(A) Change packaging to `war` in `pom.xml` and extend `SpringBootServletInitializer`

(B) Rename extension manually

(C) Add Tomcat runtime manually to target

(D) Not supported in Spring Boot

Q113. What is the purpose of Spring Data JPA `@NoRepositoryBean`?

(A) Prevents repository interface from being instantiated directly as a bean

(B) Disables database connection

(C) Marks repository as read-only

(D) Marks broken repository

Q114. Which HTTP method is typically used to perform partial updates to an existing resource?

(A) PUT

(B) PATCH

(C) POST

(D) UPDATE

Q115. What does `@EnableAutoConfiguration` do in Spring Boot?

(A) Scans components automatically

(B) Tells Spring Boot to start adding beans based on classpath settings

(C) Enables transactional management

(D) Enables Security

Q116. What is the purpose of `@Scheduled` annotation in Spring?

(A) To schedule background asynchronous tasks using cron or fixed rates

(B) To delay application startup

(C) To schedule database backups

(D) To clean session memory

Q117. Which response HTTP status code is returned when authentication credentials are missing or invalid?

(A) 400 Bad Request

(B) 401 Unauthorized

(C) 403 Forbidden

(D) 405 Method Not Allowed

Q118. Which JDBC `ResultSet` method is used to move cursor to next record?

(A) `rs.first()`

(B) `rs.next()`

(C) `rs.forward()`

(D) `rs.move()`

Q119. In REST API design, what does statelessness imply?

(A) Server stores user session state across requests

(B) Each HTTP request contains all necessary data to process it

(C) Database cannot store data

(D) Cookies are mandatory

Q120. What is the default lifecycle scope of a Servlet instance managed by container?

(A) Prototype (one instance per request)

(B) Singleton (one instance shared across requests)

(C) Session scope

(D) Thread scope

Answer Key (For Instructor / Reference)

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	18	B	35	B	52	B	69	B	86	A
2	C	19	B	36	A	53	B	70	C	87	A
3	B	20	B	37	B	54	B	71	B	88	B
4	B	21	B	38	A	55	A	72	B	89	B
5	B	22	C	39	C	56	C	73	B	90	A
6	C	23	C	40	C	57	B	74	A	91	D
7	C	24	B	41	B	58	B	75	B	92	B
8	B	25	C	42	C	59	B	76	C	93	A
9	B	26	C	43	B	60	B	77	A	94	B
10	A	27	B	44	B	61	A	78	A	95	A
11	B	28	C	45	B	62	A	79	C	96	B
12	B	29	B	46	B	63	D	80	B	97	B
13	C	30	A	47	B	64	B	81	C	98	B
14	B	31	B	48	C	65	B	82	B	99	B
15	C	32	C	49	C	66	B	83	A	100	B
16	A	33	B	50	B	67	A	84	B		
17	B	34	B	51	B	68	B	85	B		